

Chapter 1: The four core principles of economics

PROFESSOR MIKAL SKUTERUD

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UNIVERSITY OF WATERLOO

What is an economy?



Lord of the Flies by William Golding
published in 1954.

- Airplane crashes on a remote Pacific island. Survivors are a group of boys aged 6 to 12.
- Scarce resources available on the island for survival.
- What to produce? How to produce? Who does what? Who gets what?

The principled approach to economics

- **Economics** is the study of **economies** or **economic systems**. **Economists** are people who do economics.
- 250 years of economic thought has led to some core ideas or “principles of economics.”
- The goal of this course is not to memorize definitions or facts about the economy. The goal is to learn how to “do economics” by thinking like an economist.
- At the core of economics are **four principles**. As you’ll see, these principles are highly relevant to your everyday life. They will help you to make better decisions in your personal and professional life.
- The building blocks (“atoms”) of economics are individual decisions. In understanding how people make **decisions**, we begin to understand what gets produced, who produces it, and who gets what.

1. Cost-Benefit Principle

- Only pursue options when the **benefit** is at least as big as the **cost**.
- But how to compare costs and benefits that don't have explicit dollar values?
 - ➡ What's the most that you'd be willing to pay to obtain the benefit or to avoid the cost?
- But not everything is about money.
- But sometimes the benefit goes to someone else.
- But costs and benefits are sometimes uncertain.
- Costs and benefits are the **economic incentives** that determine how people make decisions.

Economic surplus

- The difference between the benefit you receive and the cost from you incur from a decision is the **economic surplus**. It is a measure of how much your choice has raised your economic well-being.
- Every time we buy something, we receive some economic surplus, but what about the seller of the good? The selling price is a benefit to the seller. If the cost of producing the good is less than the selling price, they also receive a surplus. Both the **buyer** and **seller** are better off.
- This **mutual gain from voluntary exchange** lies at the heart of all economic transactions. Economic transactions are not “zero-sum games.”

2. Opportunity Cost Principle

- The true cost of something is what you must give up to get it. What you give up is the next best alternative. This is the **opportunity cost** of a choice.
- Sometimes the opportunity cost of a choice is much bigger than the out-of-pocket money cost (“financial cost”).



What is the cost of an hour spent gaming?

The biggest cost isn't the gaming hardware and software. The biggest cost is the loss of time that could have been spent doing something else.

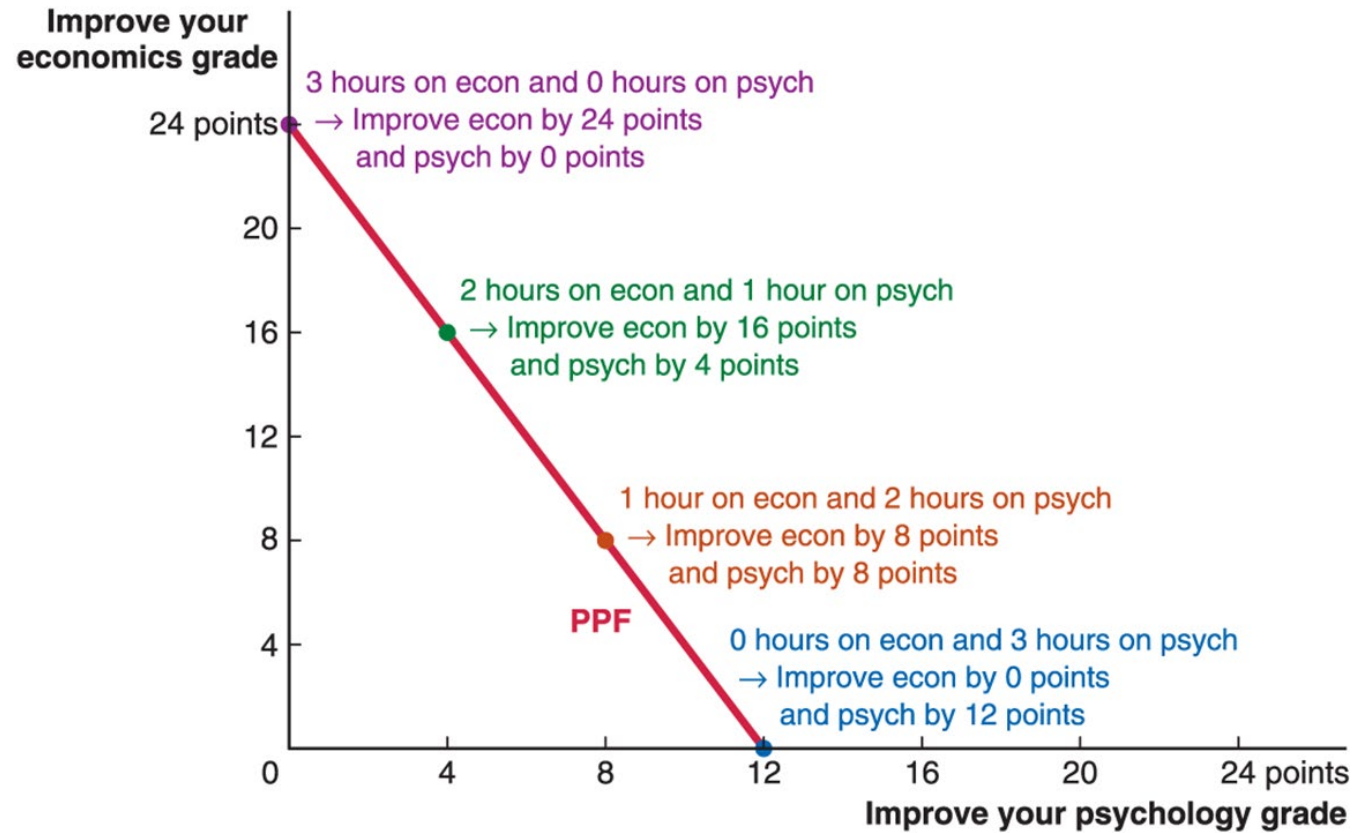
2. Opportunity Cost Principle

- What is the cost of attending today's ECON 101 lecture? Full-time tuition is \$4,500 per term for Ontario students in Arts. With 5 courses, that's \$900 per course. With 24 lectures per course, that's \$37.50 per lecture.
- Should this **out-of-pocket cost** affect your decision to attend lectures? No, because you must pay it whether you attend lectures or not. It's a **sunk cost**. The cost that matters is the 80 minutes you could have been doing something else, such as earning a wage at a job.
- Choices require resources, whether it's your income, time, or energy. But these resources are scarce. The **scarcity** of resources means that whenever we choose to do something, we are necessarily giving up ("foregoing") doing something else.
- When asking yourself whether to make a particular choice, always include "or" followed by the next best alternative in your question.

Economic phenomena consistent with the opportunity cost principle

1. Most Canadians have romantic relationships during their teen years, but why do so few get married during their teen years?
2. Why are there fewer stay-at-home moms today than 60 years ago?
3. Why are more Canadians born in September and October than in February and March?
4. Why are terminally ill people more willing to take unproven experimental drugs?

Production possibilities frontier (PPF)



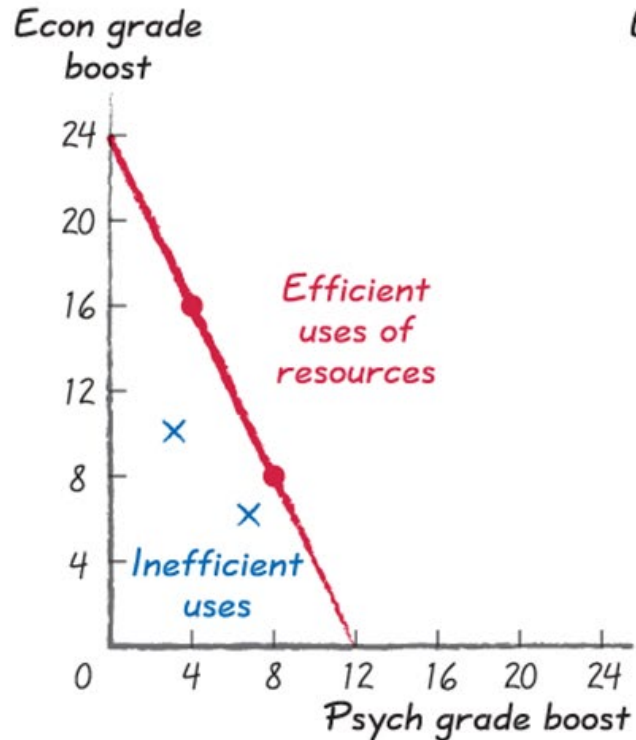
You have 3 hours per night to study economics or psychology. Each hour spent studying increases your grade but by more in economics than psychology.

The **PPF** shows the possible improvements in your grades. Notice that it is impossible to improve one grade without reducing the other.

What is the opportunity cost of a one percentage point increase in your psychology grade?

Features of PPFs

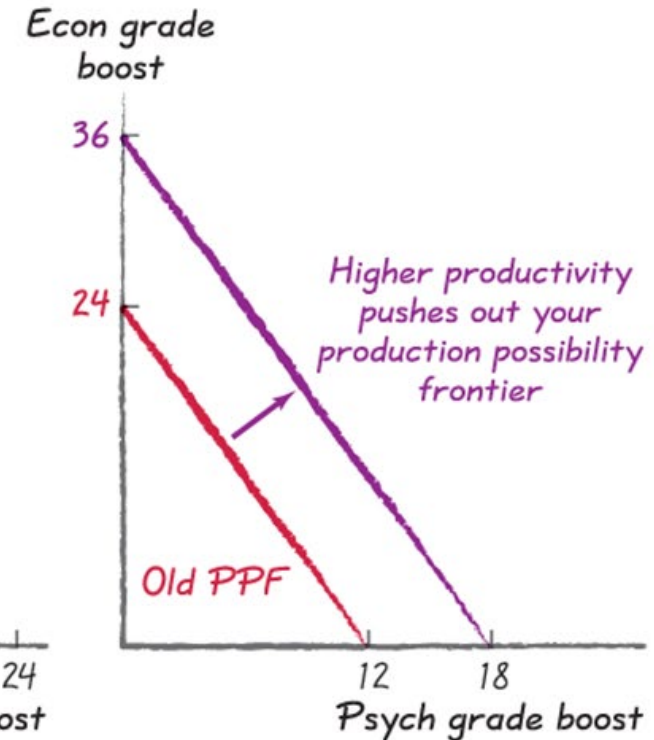
It's a frontier.



It illustrates your opportunity costs.

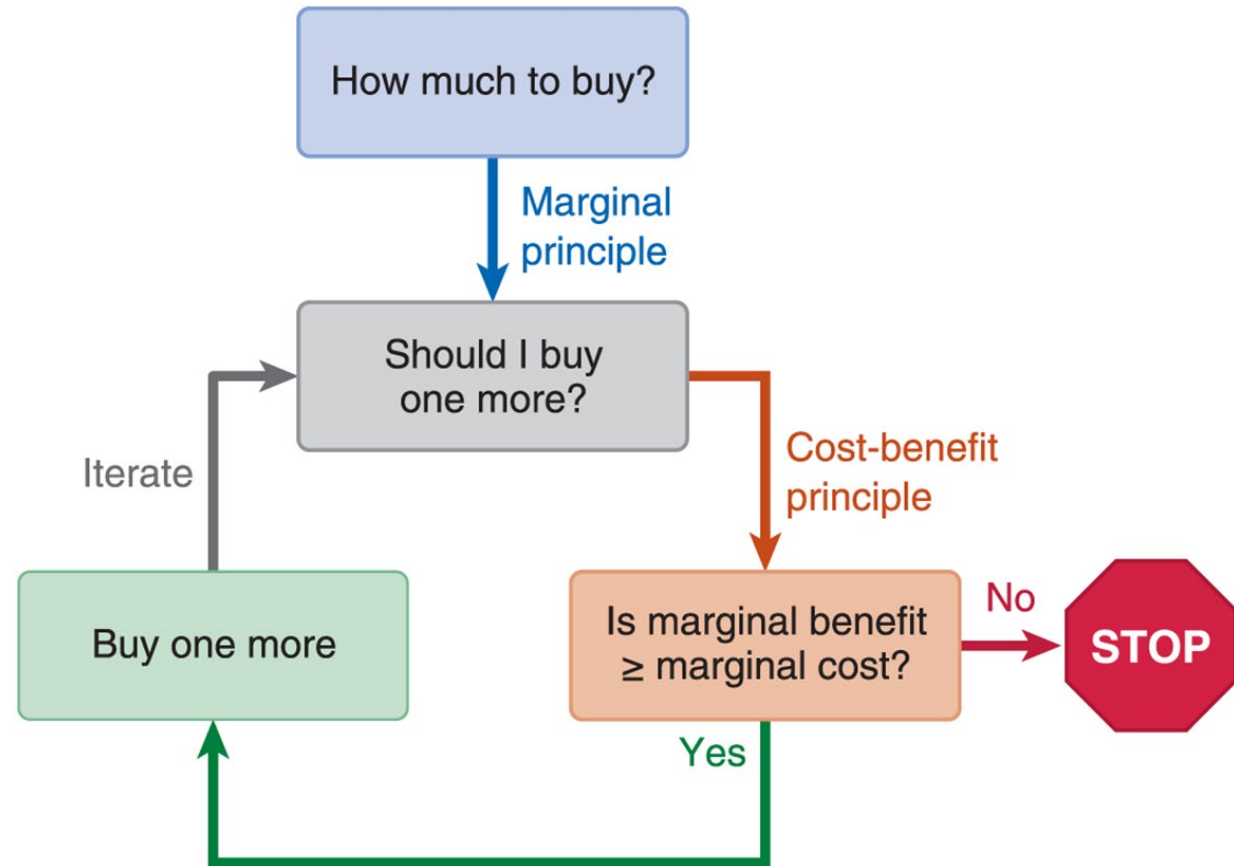


Greater productivity pushes out the frontier.



3. The Marginal Principle

- Up to now, we have only considered binary choices (either/or choices). But many choices in life involve decisions about “how many” or “how much”?
- The **marginal principle** says decisions about quantities are best made incrementally. We should break them down into a series of **marginal decisions** comparing the incremental benefits and costs of buying one more.



The rational rule

The **rational rule** says we should continue doing something as long as the **marginal benefit** of doing a little more exceeds the **marginal cost**. Following the rational rule ensures that economic surplus is maximized.

Number of workers	Meals served	Total benefits (Revenue = \$25 × number of meals)	Total costs (\$10 per meal food costs + \$300 per waiter + \$500 rent + \$1,000 for Nerida's time)
2	160	\$4,000	\$3,700
3	210	\$5,250	\$4,500
4	250	\$6,250	\$5,200
5	280	\$7,000	\$5,800
6	300	\$7,500	\$6,300
7	310	\$7,750	\$6,700

The rational rule

The **rational rule** says we should continue doing something as long as the **marginal benefit** of doing a little more exceeds the **marginal cost**. Following the rational rule ensures that economic surplus is maximized.

Number of workers	Meals served	Total benefits (Revenue = \$25 × number of meals)	Marginal benefit (Change in total benefit from hiring an extra waiter)	Total costs (\$10 per meal food costs + \$300 per waiter + \$500 rent + \$1,000 for Nerida's time)	Marginal cost (Change in total cost from hiring an extra waiter)
2	160	\$4,000		\$3,700	
3	210	\$5,250	\$1,250	\$4,500	\$800
4	250	\$6,250	\$1,000	\$5,200	\$700
5	280	\$7,000	\$750	\$5,800	\$600
6	300	\$7,500	\$500	\$6,300	\$500
7	310	\$7,750	\$250	\$6,700	\$400

Marginal benefit equals marginal cost

The rational rule

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Number of workers	Meals served	Total benefits (Revenue = \$25 × number of meals)	Marginal benefit (Change in total benefit from hiring an extra waiter)	Total costs ((\$10 per meal food costs + \$300 per waiter + \$500 rent + \$1,000 for Nerida's time)	Marginal cost (Change in total cost from hiring an extra waiter)	Profit or economic surplus (Total benefits less total costs)
2	160	\$4,000		\$3,700		\$300
3	210	\$5,250	\$1,250	\$4,500	\$800	\$750
4	250	\$6,250	\$1,000	\$5,200	\$700	\$1,050
5	280	\$7,000	\$750	\$5,800	\$600	\$1,200
6	300	\$7,500	\$500	\$6,300	\$500	\$1,200
7	310	\$7,750	\$250	\$6,700	\$400	\$1,050

Marginal benefit equals marginal cost

Maximum profit

4. The Interdependence Principle

- Your best choice in any situation doesn't happen in a vacuum but depends on:
 1. Your other choices
 - You have limited resources (income, time, attention, productive/cognitive capacity, wealth) so your choice in any situation affects the availability of resources in other situations.
 2. The choices of other decision makers
 - The resources available to you depend on the resources used by others. Scarcity results in competition between economic agents (people, business, governments) which affects your decisions.
 3. Dependencies between markets
 - Your choice in any market (labour, housing, credit) depends on what is happening in other markets.
 4. Dependencies over time
 - Your choice today is affected by your choices yesterday and your expected future choices.